

Wave Equilibrium Hypothesis: The Universe as a Balanced State of Zero

Author: Sergey Skrynnik

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Abstract

This work proposes the **Wave Equilibrium Hypothesis**, describing the Universe as a system emerging from a state of zero balance. The fundamental form of existence within such a state is the wave — an oscillatory process in which positive and negative phases mutually compensate, maintaining the total zero. The sine wave is interpreted as the projection of rotation — the simplest expression of equilibrium and the rhythm of the zero state.

Each new dimension arises not beyond the previous one but as a nested rotation around the same center. Rotational velocity defines the base frequency, and **frequency itself is the primary form of existence**. The entire structure of reality can thus be described as a **fractal-resonant system**, where matter, energy, and consciousness represent different levels of frequency interaction.

The **centripetal force** is interpreted as a universal principle underlying all forms of interaction — from elementary particles to galaxies, from matter to consciousness. On the physical level, it manifests as the force of equilibrium; on the level of perception, it manifests as **Love**, the inner resonance that unites all forms of being. Hence, **Love and the centripetal force** are two aspects of the same universal law that maintains the harmony of the Universe.

Keywords:

wave equilibrium, zero state, origin of the universe, sinusoidal symmetry, self-balancing system, matter as standing wave, harmonic creation

1. Introduction

Modern physical and philosophical approaches increasingly assume that the Universe could have emerged from “nothing.” If “nothing” is viewed not as absolute emptiness but as a *balanced zero state*, the question arises: what form can spontaneously appear from this state without breaking its balance? The answer may be a **wave**.

2. The Wave as a Form of Equilibrium

A wave is a dynamic form in which every deviation is compensated by its opposite. The positive and negative phases mutually balance each other, and the total value over time remains zero. Thus, a wave can **exist within zero, embodying balance** without requiring an external source of energy.

If the initial state of the world was a perfect zero, the only possible motion that preserves it is **oscillation**. The Universe may have begun not with an explosion, but with a **fluctuation of equilibrium**.

3. Why the Sine Wave

Any oscillation can be represented as a sum of sinusoidal components. The sine wave is the most stable and symmetric form of oscillation. It expresses the idea of balance: its positive and negative parts fully compensate one another, maintaining the total energy of the system.

This makes the sine wave the **minimal and fundamental form of existence** capable of spontaneously arising from zero. All more complex forms are combinations of these basic harmonics.

4. From Wave to Matter

If all reality is wave-like, then matter is a **stable interference structure**. Through interaction, waves form local regions of equilibrium perceived as particles, fields, and physical objects. Thus, **matter is a stable form of wave balance**.

5. The Law of Energy Conservation as a Manifestation of Zero

Within this hypothesis, the law of conservation of energy finds a natural explanation. If the entire system is based on the mutual compensation of phases, then the total energy of everything that exists is always zero. Energy is neither created nor destroyed — it merely oscillates between positive and negative phases, between the “crest” and the “trough” of the wave. This wave symmetry forms the basis of the observed conservation law.

6. Connection with Quantum Models

Quantum mechanics already describes particles as **wave functions** that include both positive and negative components. If these are interpreted as real oscillatory states of a zero field, the Universe can be viewed as a **wave of zero**, existing in complete energetic equilibrium.

7. Geometric Nature of the Wave: the Link Between the Sine and the Circle

The sine wave is not an abstract form; its origin is deeply **geometric**. It arises as the **projection of uniform circular motion** onto one of the coordinate axes. When a point moves around a circle with constant angular velocity, its coordinates are described by:

$$x = A \cos(\omega t), \quad y = A \sin(\omega t)$$

Considering only one coordinate (for example, y), we obtain the ordinary sine wave. This means that any wave is a **partial manifestation of rotational motion**, and the wave nature of matter is a consequence of an underlying cyclic process.

8. Circle, Sphere, and Dimensions

A circle is a projection of a sphere onto a lower dimension. If a sine wave is a projection of rotation, then perhaps the entire observed linear structure of spacetime is merely a projection of a **multidimensional rotational state**.

That is why **spherical geometry and dimensionality** are closely related:

- the point — zero dimension, balance, zero,

- the line — directly the sine wave, change, time,
- the circle — plane, rotation around a center,
- the sphere — rotation around a line passing through the center.

A wave can be regarded as a **temporal projection of rotational motion** occurring in a higher space. Such an understanding unites geometry (the circle), the wave (the sine), and time (the period) into a single system.

9. Mechanism of Dimensional Generation

A sphere represents the completed form of equilibrium within its dimension. Every point on its surface is at the same distance from the center, forming a state of perfect symmetry. However, the emergence of a new dimension is not associated with the appearance of a new center. The center remains the same — the zero point from which all rotation arises. What changes is not the location of the center, but the **nature of motion** — its embedding and relative velocity.

Geometrically, this process can be seen as follows:

1. A point is balance; it is the source of change.
2. The birth of frequency — Time — is a characteristic of change, a sine wave. For the point, it is circular — increase and decrease.
3. Frequency can only increase. The increase of frequency relative to the original one, when viewed as the initial balance, leads to the understanding of fractality — an increase of frequency by a frequency factor. At the same time, our more familiar perception is geometric — rotation — which requires accounting for the coefficient 2π — the birth of the circle. Thus, the increase actually occurs not exactly by a frequency factor, but by 2π times the frequency. In this case, the initial characteristic/frequency can now be viewed as a line — the arrow of time — while the second frequency range gives rise to the understanding of a plane relative to time and the original point/balance — the circle. Thus, the birth of the plane is directly connected to balance and time — two links, two coordinates.
4. Further increase in frequency leads to the emergence of volume in space and the appearance of the concept of mass. The circular rotation of the plane's dimension now turns into a line for the next process. Here, the frequency increases by 2π times relative to the plane's frequency. In this way, the mass connection is formed through three links to the original balance, the point — three coordinates — and rotation relative to the plane marks the birth of mass.
5. To understand how the next dimension arises, one must imagine a new rotation relative to the already existing rotation. At this level, our three-dimensional perception reaches its limit: we are capable of imagining rotation in three mutually perpendicular planes, but we cannot visually perceive the fourth axis — it manifests only as a change in frequency or speed. We can observe it only indirectly — as black holes.

Each subsequent dimension does not add an external axis but **increases the level of relative rotational speed**, and therefore — raises the **frequency**. If the first rotation has a rate of change c , then the second will relate to the initial rate as c^2 , the third as c^4 , and the fourth as c^8 . Thus, a hierarchy is formed — a fractal structure of frequencies, in which each new dimension is the result of the acceleration of the previous one.

Essentially, there exists a **certain fundamental frequency** for us. All forms of existence are interactions of waves of different frequencies. The world can be viewed as a system in which each structure is the result of resonant connections between these frequencies. Equilibrium arises when waves of different levels of “rotation” are in harmonic proportion, and it is precisely this resonance that holds together and forms the overall picture of the observable Universe in a state of unified balance.

Thus, dimensions are not static coordinates but a **hierarchy of nested frequency rotations connected through resonance**. Each new dimension is a new level of relative motion with an unchanging center — zero — which maintains the equilibrium of the entire system.

Formally, this process can be expressed as:

$$D_{n+1} = R(D_n)$$

where D_n is the form of the current dimension,
and R — the operation of embedded rotation (increase of frequency) around the same center — zero.

Thus manifests the **principle of fractal rotation of zero** — the gradual self-unfolding of equilibrium through frequency levels of motion. All forms of existence, from particles to galaxies, are resonant expressions of this single universal process.

Relation to Human Systems and Perception

This principle applies not only to geometry but also to any system possessing internal structure — including the human being. Consciousness, like matter, can be regarded as a form of wave equilibrium capable of transitioning to a higher level by increasing the speed of perception, that is, its frequency. In this way, the expansion of awareness occurs — the birth of a new dimension within perception.

10. Frequency as the Primary Form of Existence

Considering the structure of the Universe as a system emerging from the zero state, one can make an essential clarification: the primary element of this system is not motion or geometry, but **frequency**.

Frequency expresses the rhythm of zero equilibrium — the mutual oscillation of the positive and negative aspects of being. It does not require extension, form, or spatial localization; it exists as a **relation**, as the very possibility of transition between states.

Rotation and motion are projections of frequency into dimensions where form becomes observable. What we call rotation is the **geometrization of frequency**, and the sine wave is the visible imprint of this process within the plane of perception.

Thus, everything we perceive as form, motion, or matter is a **manifestation of interacting frequencies** bound through resonance.

Space emerges where frequencies interact, time — where their sequence appears, and energy — where their difference manifests.

Therefore, **frequency is the primary essence of being**, and the entire Universe is a process of maintaining harmonic equilibrium among frequency states. When complete correspondence between them is achieved, the system appears as rest. When a phase difference arises — it appears as motion and life.

Conclusions

1. The Universe can be viewed as a system emerging from a state of zero equilibrium, where the wave is the fundamental form of existence.
2. Rotation and sinusoidal forms are projections of frequency, making the structure of zero perceivable in space.
3. Each new dimension arises not through the addition of a new axis, but through an increase in rotational frequency — through nested levels of resonance.
4. Everything we are able to observe is defined by the fundamental frequency of changes, which determines the basic rate of interaction, and all forms of matter and energy are the result of the interaction of different frequencies.
5. Geometry is not the cause but the consequence of frequency interactions — the frozen shape of rhythm.
6. System equilibrium is achieved through harmonic resonance, which on the physical level appears as stability, and on the philosophical level — as **Love**, the unifying and centering force.
7. The conservation of energy is a **result of the symmetry of the zero state**.

Formally expressed as:

$$A \sin(\omega t) + (-A \sin(\omega t)) = 0$$

Frequency is primary, form is its consequence, and resonance is the language through which the Universe expresses its equilibrium.

Philosophical Interpretation of the Centripetal Principle

If rotation is considered the primary act of motion, then it is the first building block of interaction on the physical level. Every form of movement tends toward the center — and in this, the centripetal force manifests itself, which is, in essence, the only fundamental force in the Universe. All other forces can be regarded as particular manifestations or decompositions of this principle — similar to how a complex wave can be represented by a Fourier series as the sum of simple harmonics.

The centripetal force maintains the equilibrium of all levels — from the elementary particle to the galaxy, from physical motion to the processes of consciousness. It binds everything that exists into a single structure, preventing the system from disintegrating.

At the level of consciousness, this same principle manifests as the force of Love — an inner attraction, a striving for unity, a desire to return everything to the center from which all has arisen. Love is a reflection of the very same law that, in physics, is expressed as the centripetal force.

Thus, rotation is the form of expression of frequency, the centripetal force is the manifestation of equilibrium, and Love is the conscious side of this law.

On the physical level, it holds matter together; on the spiritual level, it unites consciousness; and as a whole, it sustains the balance of the Universe as a single entirety.

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